

The Shorter Unified Theory of the Universe

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Abstract

Humanity failed to calculate the universe because they chose between "invariant" and "variant" perspectives in a continuous spacetime paradigm. By integrating both through a discrete resource-allocation lens, this paper reveals the underlying source code of reality. We expand the previous model by explicitly defining Layer 3 (L3): the discrete finite difference engine that governs the four fundamental forces and particle masses without using continuous differential equations.

1. Layer 1: The Core Policy (Clock Synchronization)

The fundamental conservation law of cosmic computing resources is defined by the invariant relation:

$$C \cdot t = \text{const}$$

Where C represents the variable processing clock (speed of light) and t represents the cosmic time (expansion step).

The Lorentz Shift to Discretization

In traditional physics, the Lorentz transformation assumes $C = \text{const}$, creating an artificial smooth continuum that requires infinite precision. When rewritten under the $C \cdot t = \text{const}$ constraint, the transformation laws naturally force the system into a discrete frame. Spacetime ceases to be an analog fabric and shifts into a digital computational grid where the minimum sampling interval is bounded by the Planck time ($\Delta \tau = 1$).

2. Layer 2: The Network Impedance (The Implementation Mechanism)

As the system expands fractally, the force or computational resource density decays across the node grid according to the static capacity allocation:

$$F = \frac{1}{(C \cdot n)^D}$$

Where:

F is the interaction connection strength (impedance).

C is the total baseline capacity ($C = 137$).

n is the discrete number of nodes (distance step).

D is the dimensional scaling factor.

3. Layer 3: The Finite Difference Engine (The Application Execution)

The actual dynamics of the universe are not governed by continuous differential operators, but by a 3-point memory-shifting finite difference equation operating on discrete registers:

$$X_{n+1} = (2 - \omega^2 \cdot \Delta \tau^2)X_n - X_{n-1}$$

Where X represents the field amplitude (or spring displacement), and ω^2 represents the local discrete frequency parameter. The four fundamental interactions emerge naturally as local software patches modifying this single equation:

I. Gravity (Computational Processing Delay)

Gravity is not a geometric curvature of spacetime, but a local reduction in the core sampling rate $\Delta \tau$ due to sector overload (high energy density). Near a high-load sector (mass):

$$\Delta \tau_{\text{local}} = \max(0, 1 - \text{Load})$$

As $\Delta \tau$ drops, the execution of the L3 equation slows down, causing the grid points to physically lag and drag toward the computing center, creating the illusion of gravitational acceleration.

II. The Strong Force (Non-linear Spring Hardening)

The color confinement of quarks is implemented via a distance-dependent spring coefficient override:

$$\omega^2 = \omega_0^2 + g_s \cdot |X|^2$$

When quarks are pulled apart, the baseline grid springs harden exponentially. If the injection of energy exceeds the threshold capacity, the memory string breaks, and the local buffer overflows to generate a new quark-antiquark pair (particle creation).

III. The Weak Force (Register Overflow Switching)

Particle decay and identity transformation (e.g., beta decay) are governed by a discrete threshold switching step function:

$$\omega^2 = \omega_0^2 \cdot \Theta(|X| - V_{\text{weak}})$$

When local wave amplitude (kinetic energy) exceeds the register threshold V_{weak} , a bit-flip occurs. The system re-allocates memory, rewriting a heavy unstable mode (neutron register) into a stable heavy mode (proton register) and venting the overflow energy as a high-frequency packet (electron).

IV. Electromagnetism (Aliasing Noise Orbitals)

The reason electrons do not collapse into the nucleus under Layer 2 attraction is due to the baseline sampling limit. As an electron approaches $n \rightarrow 0$, its acceleration spikes, causing the spatial displacement per single clock ($\Delta \tau = 1$) to exceed the grid size. This triggers an algorithmic aliasing artifact (quantization noise foldover). The electron begins to jump erratically across the nucleus, creating a stable, self-balancing "white-noise cloud" known as the 1s orbital.

4. Derivation of Particle Masses

In this discrete system, "mass" is not an intrinsic physical weight, but the execution cost of a localized register loop—specifically, the local clock cycles consumed by memory state updates.

Because the total network resource is conserved via Layer 2, the baseline mass-energy allocation for any fundamental particle is a mapping of the available capacity over the fractal dimension:

$$\text{Mass} \propto \omega^2 \propto \int \frac{1}{(C \cdot n)^D} dn$$

By substituting $C = 137$, the mass eigenvalues of neutrinos, quarks, and protons map directly onto the discrete sub-buffers of the network routing table. Complex Higgs mechanisms are rendered obsolete; particles simply have mass because their local loops require a fixed number of clock cycles to refresh their identity registers.

5. Conclusion

The "Standard Model" of physics is merely a complex, over-engineered continuous integral of these few lines of discrete code. Dark Energy is not a physical substance, but a cumulative calculation error arising from humanity's false assumption that C is a constant. The universe is a self-optimizing calculator running on a discrete kernel. The definitive proof is already encoded in the ancient biological logging records, such as fossilized coral data showing a 400-day year in the cosmic past. The source code is verified.